18

CHAPTER

HYDRAULIC MACHINES — TURBINES

▶ 18.1 INTRODUCTION

Hydraulic machines are defined as those machines which convert either hydraulic energy (energy possessed by water) into mechanical energy (which is further converted into electrical energy) or mechanical energy into hydraulic energy. The hydraulic machines, which convert the hydraulic energy into mechanical energy, are called *turbines* while the hydraulic machines which convert the mechanical energy into hydraulic energy are called *pumps*. Thus the study of hydraulic machines consists of study of turbines and pumps. Turbines consists of mainly study of Pelton turbine, Francis Turbine and Kaplan Turbine while pumps consist of study of centrifugal pump and reciprocating pumps.

▶ 18.2 TURBINES

Turbines are defined as the hydraulic machines which convert hydraulic energy into mechanical energy. This mechanical energy is used in running an electric generator which is directly coupled to the shaft of the turbine. Thus the mechanical energy is converted into electrical energy. The electric power which is obtained from the hydraulic energy (energy of water) is known as *Hydroelectric power*. At present the generation of hydroelectric power is the cheapest as compared by the power generated by other sources such as oil, coal etc.

▶ 18.3 GENERAL LAYOUT OF A HYDROELECTRIC POWER PLANT

Fig. 18.1 shows a general layout of a hydroelectric power plant which consists of :

(*i*) A dam constructed across a river to store water.

(*ii*) Pipes of large diameters called penstocks, which carry water under pressure from the storage reservoir to the turbines. These pipes are made of steel or reinforced concrete.

(*iii*) Turbines having different types of vanes fitted to the wheels.

(*iv*) Tail race, which is a channel which carries water away from the turbines after the water has worked on the turbines. The surface of water in the tail race channel is also known as tail race.

▶ 18.4 DEFINITIONS OF HEADS AND EFFICIENCIES OF A TURBINE

1. Gross Head. The difference between the head race level and tail race level when no water is flowing is known as Gross Head. It is denoted by 'H_g' in Fig. 18.1.

2. Net Head. It is also called effective head and is defined as the head available at the inlet of the turbine. When water is flowing from head race to the turbine, a loss of head due to friction between the water and penstocks occurs. Though there are other losses also such as loss due to bend, pipe fittings, loss at the entrance of penstock etc., yet they are having small magnitude as compared to head loss due to friction. If 'h_f' is the head loss due to friction between penstocks and water then net heat on turbine is given by

$$H = H_g - h_f \qquad ...(18.1)$$

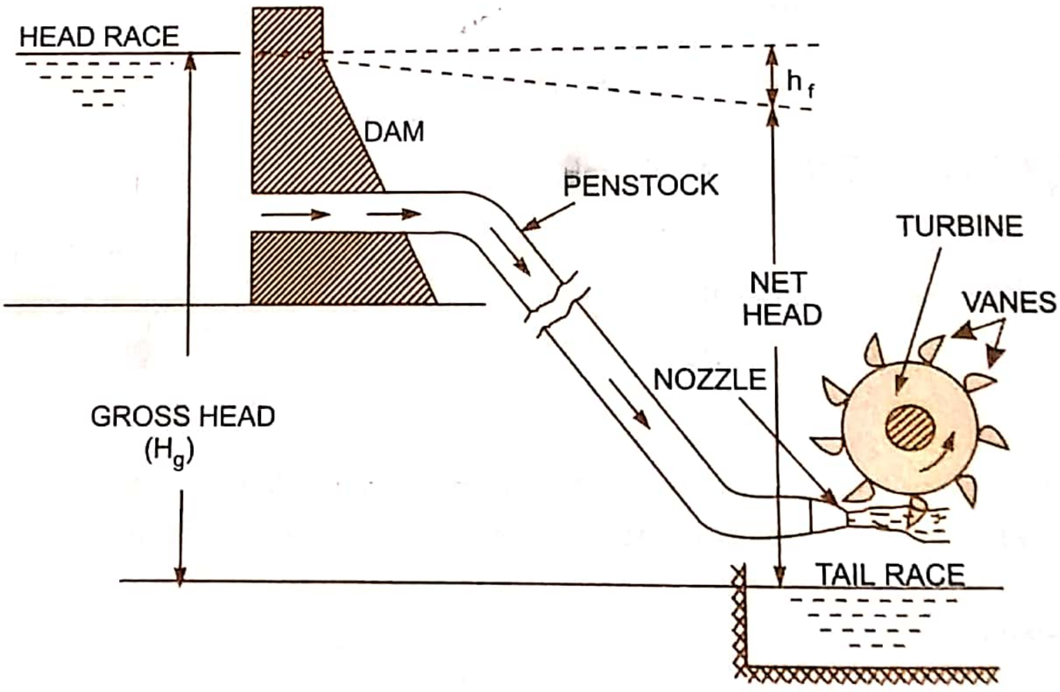

Fig. 18.1 *Layout of a hydroelectric power plant.*

▶ 18.5 CLASSIFICATION OF HYDRAULIC TURBINES

The hydraulic turbines are classified according to the type of energy available at the inlet of the turbine, direction of flow through the vanes, head at the inlet of the turbine and specific speed of the turbines. Thus the following are the important classifications of the turbines :

1. According to the type of energy at inlet :
   (*a*) Impulse turbine, and (*b*) Reaction turbine.
2. According to the direction of flow through runner :
   (*a*) Tangential flow turbine, (*b*) Radial flow turbine,
   (*c*) Axial flow turbine, and (*d*) Mixed flow turbine.
3. According to the head at the inlet of turbine :
   (*a*) High head turbine, (*b*) Medium head turbine, and
   (*c*) Low head turbine.
4. According to the specific speed of the turbine :
   (*a*) Low specific speed turbine, (*b*) Medium specific speed turbine, and
   (*c*) High specific speed turbine.

If at the inlet of the turbine, the energy available is only kinetic energy, the turbine is known as **impulse turbine.** As the water flows over the vanes, the pressure is atmospheric from inlet to outlet of the turbine. If at the inlet of the turbine, the water possesses kinetic energy as well as pressure energy, the turbine is known as **reaction turbine**. As the waters flows through the runner, the water is under pressure and the pressure energy goes on changing into kinetic energy. The runner is completely enclosed in an air-tight casing and the runner and casing is completely full of water.

If the water flows along the tangent of the runner, the turbine is known as **tangential flow turbine.** If the water flows in the radial direction through the runner, the turbine is called **radial flow turbine.** If the water flows from outwards to inwards, radially, the turbine is known as **inward** radial flow turbine, on the other hand, if water flows radially from inwards to outwards, the turbine is known as **outward** radial flow turbine. If the water flows through the runner along the direction parallel to the axis of rotation of the runner, the turbine is called **axial flow** turbine. If the water flows through the runner in the radial direction but leaves in the direction parallel to axis of rotation of the runner, the turbine is called **mixed flow** turbine.

▶ 18.6 PELTON WHEEL (OR TURBINE)

The Pelton wheel or Pelton turbine is a tangential flow impulse turbine. The water strikes the bucket along the tangent of the runner. The energy available at the inlet of the turbine is only kinetic energy. The pressure at the inlet and outlet of the turbine is atmospheric. This turbine is used for high heads and is named after L.A. Pelton, an American Engineer.

Fig. 18.1 shows the layout of a hydroelectric power plant in which the turbine is Pelton wheel . The water from the reservoir flows through the penstocks at the outlet of which a nozzle is fitted. The nozzle increases the kinetic energy of the water flowing through the penstock. At the outlet of the nozzle, the water comes out in the form of a jet and strikes the buckets (vanes) of the runner. The main parts of the Pelton turbine are :

1. Nozzle and flow regulating arrangement (spear),
2. Runner and buckets,
3. Casing, and
4. Breaking jet.

1. Nozzle and Flow Regulating Arrangement. The amount of water striking the buckets (vanes) of the runner is controlled by providing a spear in the nozzle as shown in Fig. 18.2. The spear is a conical needle which is operated either by a hand wheel or automatically in an axial direction depending upon the size of the unit. When the spear is pushed forward into the nozzle the amount of water striking the runner is reduced. On the other hand, if the spear is pushed back, the amount of water striking the runner increases.

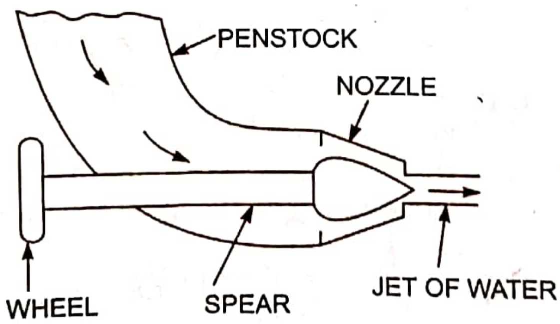

Fig. 18.2 *Nozzle with a spear to regulate flow.*

2. Runner with Buckets. Fig. 18.3 shows the runner of a Pelton wheel. It consists of a circular disc on the periphery of which a number of buckets evenly spaced are fixed. The shape of the buckets is of a double hemispherical cup or bowl. Each bucket is divided into two symmetrical parts by a dividing wall which is known as splitter.

6. Braking nozzle

Casing

The casing of the pelton wheel does not perform any hydraulic function. The function of the casing is to prevent the splashing of water striking the buckets and to lead water to the tail race. It also acts as safeguard against accidents. Pelton wheel does not require air-tight casing because it works on atmospheric pressure. The casing is made of cast iron or fabricated steel plates.

Braking nozzle

Braking nozzle is used to stop the turbine immediately during shut-down. For the purpose of stopping the turbine, the water flow through the nozzle is completely stopped by closing the nozzle by means of spear valve. But due to high inertia of the runner, it revolves for a long time continuously. In order to bring the runner to rest quickly in short time, a brake nozzle is provided which directs water on the back of buckets. This produces a braking torque on the bucket and stops the runner immediately. This jet of water is called **braking jet.**

9.4. FRANCIS TURBINE

Modern Francis turbine is a mixed flow reaction turbine, named after its design engineer. The turbine can operate under medium head and requires medium quantity of water. It is usually employed where the water head available is from 40 m to 300 m.

9.4.1. Functional Details of Major Components

Figure 9.4. shows the front view and top view of a Francis turbine .The major elements of Francis turbine are scroll casing, guide vanes, runner with vanes and draft tube.

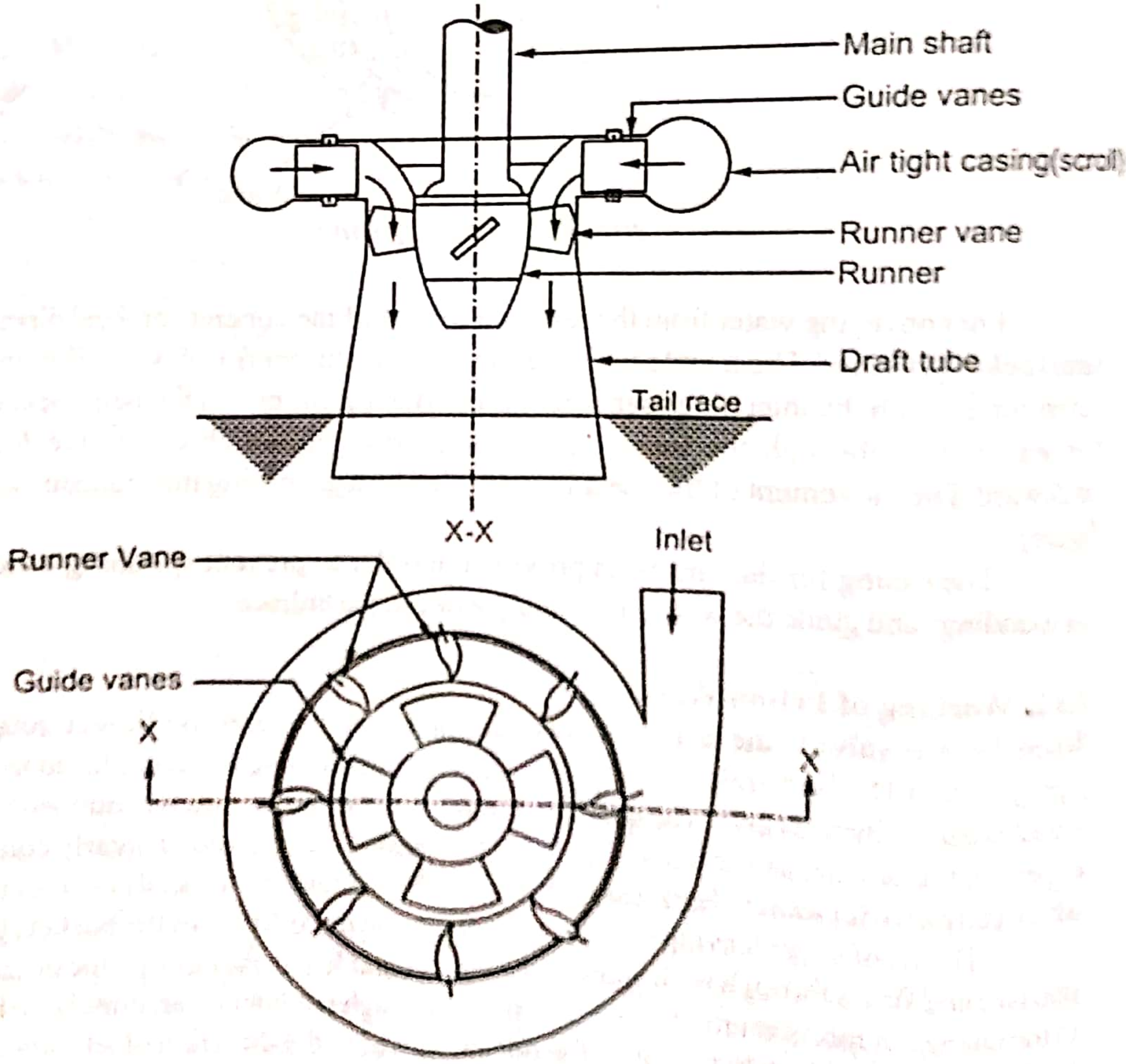

Fig. 9.4. Francis turbine

Scroll casing (also called **spiral casing**), made of cast steel or concrete is an annular channel surrounding the turbine runner. The casing is spiral in layout such that flow area decreases gradually in the direction of flow, to maintain a constant velocity. Water entering through the inlet fills the casing and enters the guide vanes equally throughout the circumference.

Guide Vanes (also called **wicket gates**) are a series of airfoil shaped fixed vanes made of cast steel, arranged inside the casing, surrounding the runner. It directs water from the casing to the runner at the required angle.

Guide wheels are rings (two numbers), in the form of a wheel, on to which the guide vanes are pivoted.

Runner of the turbine consists of a series of curved vanes arranged equiistantly around its circumference. The shape of the vanes are such that water enters radially inward and leaves axially at the inner periphery. The force on the vanes (to make it rotate) is both due to impulse caused by flow turning and due to the reaction caused by pressure difference.

The **runner** is made of cast steel or stainless steel and is keyed to the turbine shaft. The power produced by runner is transmitted to the generator through the shaft.

Draft tube is the passage between runner and the tailrace. Its cross section gradually increases towards the outlet. The outlet end is submerged into the tailrace sothat the passage is completely sealed from the surrounding atmosphere.In the figure a straight conical draft tube is shown.

The draft tube provides a negative pressure (pressure less than atmospheric) at the outlet of the runner, sothat the net head available for work is the level difference between head race and tailrace. If the draft tube is not used, then the net head available for work is the level difference between head race and the runner exit.

9.5. KAPLAN TURBINE

Kaplan turbine is an axial flow reaction turbine particularly suitable for low head (upto 30 m) and large flow rates. It is similar to a Francis turbine except for two differences. One is the shape of the runner vanes. The runner vanes of Kaplan turbine is such that water enters and leaves the vanes axially. The second difference is that the runner vanes of Francis turbine is fixed whereas in Kaplan turbine, it is movable. The blade angle can be adjusted such that the turbine can give better efficiency even at off-design conditions.

9.5.1. Major Components

The main features of Kaplan turbine is shown schematically in Fig. 9.5. All major parts of the turbine such as spiral casing, guide vanes and draft tube are similar to Francis turbine. In the figure an L-shaped draft tube is shown.

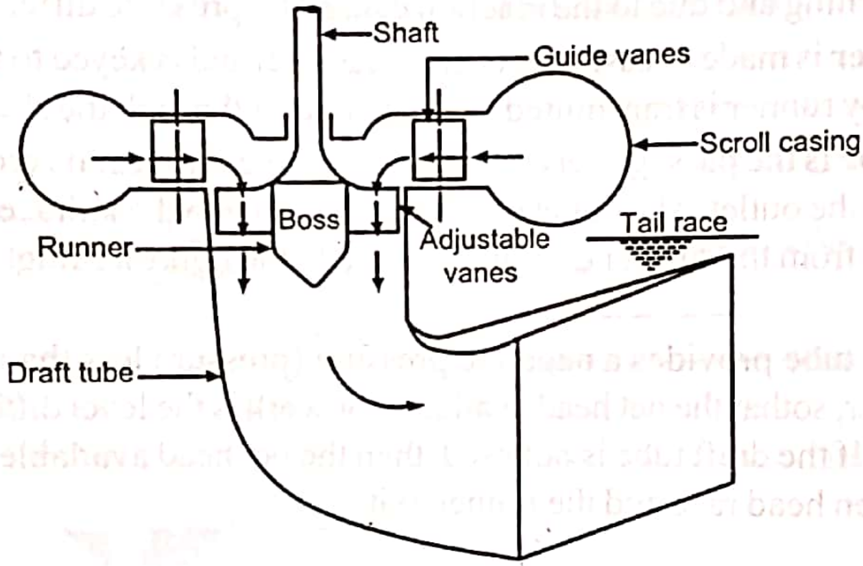

Fig. 9.5. Kaplan Turbine

The runner of the turbine is in the form of a boss (hub) and is the extension of the bottom end of the shaft. Vanes (3 to 6 in number), made of stainless steel, are mounted on the periphery of the boss. The vanes are of adjustable type, so that its angle can be changed, as required.

9.5.2. Working

Water flows into the scroll casing of the turbine and gets directed radially inward with the help of the guide vanes. A part of the available energy is converted to kinetic energy in the guide vanes and the rest remains as pressure head. The runner always flows full, since its inlet pressure is more than that at outlet. From the guide vanes water enters the runner. There is sufficient space between the guide vanes and the runner vanes, where the direction of flow changes from radial to axial. Thus water enters and leaves the runner vanes axialy.

As water passes over the shaped runner vanes, the change in momentum of water gives rise to impulsive force on the vanes. Along with that there is pressure reduction between the inlet and outlet of the vanes due to the conversion from pressure to kinetic energy. This gives rise to reaction force on the blades. Due to the two forces on the vanes, the vanes along with ~~runner~~ and shaft rotates, generating mechanical power.

8.2. CENTRIFUGAL PUMP

The basic principle of working of a centrifugal pump is a follows: When a liquid mass is made to rotate by external means, it is thrown away from the centre of rotation, by which kinetic energy is imparted to it. The kinetic energy is then converted to pressure head in the **casing** of the pump for lifting the liquid. Since the liquid is lifted by centrifugal action, the pump is called centrifugal pump.

Centrifugal pump has high output and high efficiency. Because of its simple design, it is used in almost all fields. Centrifugal pump is not used in places where high head is needed and where liquid of very high viscosity is to be pumped.

8.2.1. Major Components

The major components of a single stage centrifugal pump are shown in Fig.8.1, and are discussed below:

i) **Rotor** usually known as **Impeller:** It is the rotating element of the pump and is provided with a series of curved vanes. The impeller is mounted on a shaft which is coupled to a prime mover (mostly electric motor). The prime mover imparts mechanical energy for rotating the impeller.

ii) **Casing** around impeller: It is an airtight chamber which surrounds the impeller. It guides liquid to the impeller and from the impeller to the delivery pipe. It contains suction and discharge arrangements and supports for shaft bearings. The shape of the casing is such that the area of flow on the outer periphery of impeller gradually increases till the delivery pipe. Due to this shape, the liquid thrown out from the impeller at high velocity gets collected and allowed to flow at a lower velocity to the delivery pipe. The kinetic energy of the liquid is thus converted to pressure head.

iii) **Suction pipe:** It is the pipe through which liquid is sucked from the sump to the inlet (eye) of the impeller. At its lower end a **strainer** is usually fitted to keep out debris from entering the pump.

iv) **Foot valve:** It is a one way valve (non return valve) provided at the lower end of suction pipe. The valve is arranged such that it allows liquid to flow only upwards to the pump.

v) **Delivery pipe:** It is the pipe connecting the delivery end of the pump to the delivery tank kept at a higher level.

vi) **Delivery valve:** It is a regulating valve provided just near the pump outlet. It serves to control the flow of liquid to the delivery pipe.

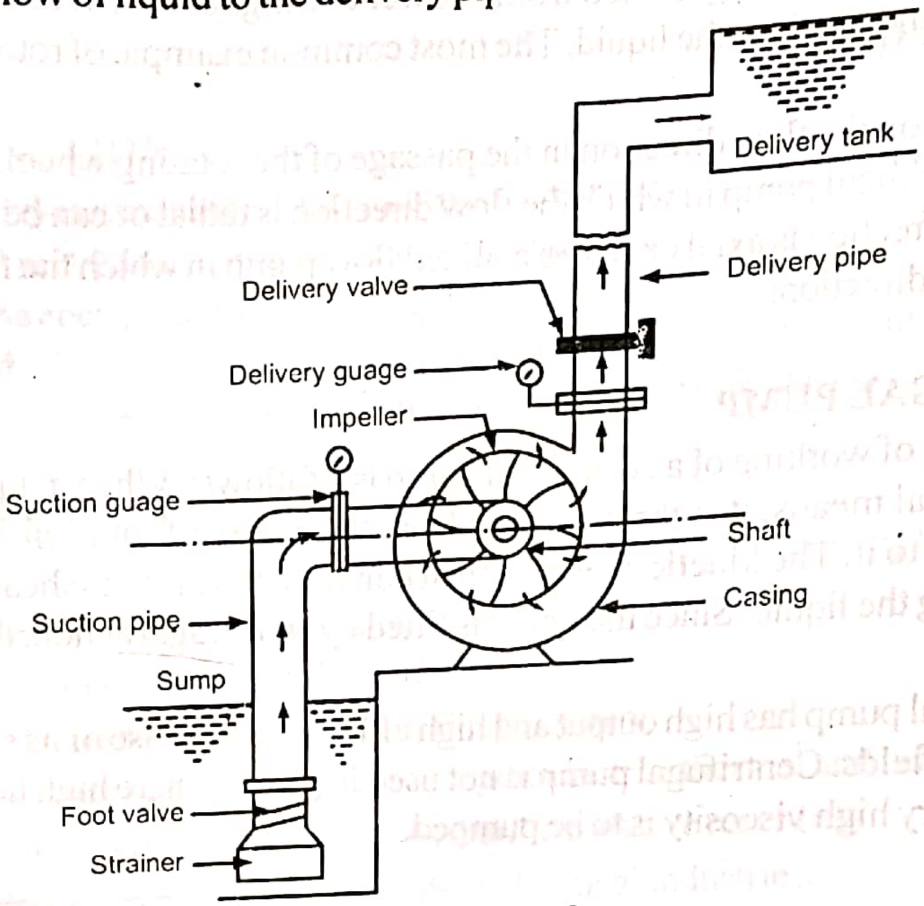

Fig. 8.1. Centrifugal pump

8.2.2. Working of Centrifugal Pump

Before starting the pump, it has to be primed. **Priming** is the process of filling the suction pipe, casing and delivery pipe upto delivery valve with the liquid to be pumped, so that no air pocket is left.

After priming, the prime mover coupled to the pump is started, keeping the delivery valve closed. This imparts rotation to the impeller. The impeller rotation imparts centrifugal force on the liquid entrapped in the impeller and throws the liquid towards the outer periphery of the impeller.

After the impeller attains normal speed, the delivery valve is opened. The liquid flows outward and leaves the vanes with higher pressure and velocity. Outward movement of liquid in the impeller creates a partial vacuum near the eye (inlet) of the impeller. Consequently, liquid from the sump is sucked in towards the impeller eye and enters through the inlet tip of impeller vanes. Thus, there is a continuous flow of liquid from the sump to the casing.

The liquid leaving the impeller vanes is at a higher pressure and velocity. The velocity head is converted to pressure head in the casing, utilising the shape of the casing. In some pumps, additional diffuser vanes having increasing area of flow are provided to effectively convert the velocity into pressure.

From the casing, liquid continuously pass to the delivery pipe and due to its pressure, liquid get lifted to the required height. From the delivery pipe, liquid is discharged to the delivery tank, kept at the required higher elevation

While stopping the pump, the delivery valve is first closed to avoid backflow from the delivery pipe and then the prime mover is stopped.

8.3. RECIPROCATING PUMP

The reciprocating pump is a positive displacement pump. It creates the required lift and pressure by displacing the liquid using a moving mechanical element called **plunger (piston)** housed inside a cylinder. Most of the reciprocating pumps have more than one cylinder for pumping. Reciprocating pumps are the best suited for places where high pressure pumping is required.

8.3.1. Major Components

Figure 8.4 shows the schematic of a single stage, single acting reciprocating pump. It primarily consist of:

1. A **plunger** (piston) that moves backward and forward in a close fitting cylinder.
2. **Suction and delivery pipes** fitted to the cylinder as shown in Fig. 8.4.
3. Two non return (one way) valves; a **suction valve** in the suction line and a **delivery valve** in the delivery line.
4. **Crank, connecting rod** and **piston rod** (crank mechanism) to convert rotary motion of the crankshaft to reciprocating motion of the piston.
5. A **prime mover** (mostly electric motor) to impart rotational power to the crankshaft.

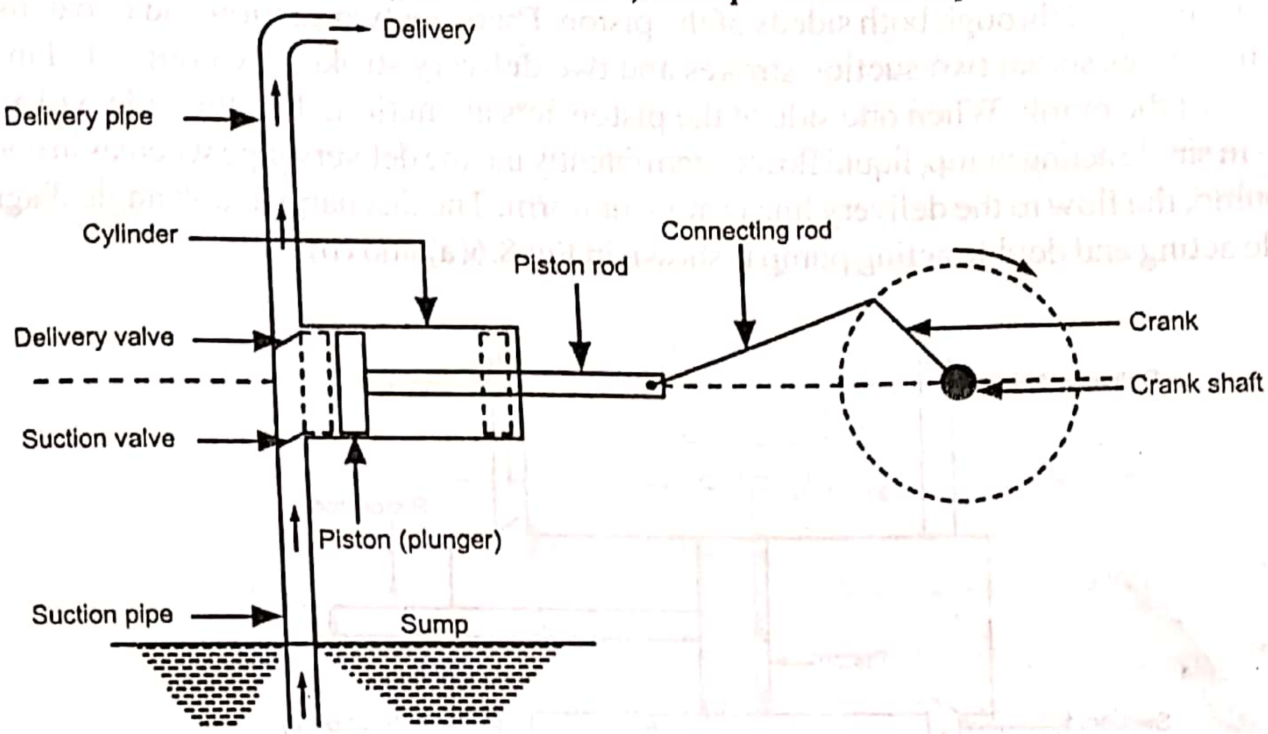

Fig. 8.4. Single acting reciprocating pump

8.3.2. Working of Reciprocating Pump

Consider a single cylinder, single acting pump as shown in Fig. 8.4. When the plunger moves towards right side (from inner dead centre to the outer dead centre) partial vacuum is developed in the left space of the cylinder. The vacuum in the region causes suction valve to open. Since atmospheric pressure is acting on the liquid surface in the sump, liquid from the sump moves up through the suction pipe and fills the left side of the cylinder. This stroke of the plunger is called **suction stroke.**

At the end of the stroke,the cylinder is completely filled with liquid from the sump.When the plunger moves towards right, pressure is built up in the cylinder. This high pressure causes the suction valve to close and the delivery valve to open. The liquid from the cylinder is now forced out through the delivery valve to the delivery pipe and raised to the required height. This stroke of the plunger is called **delivery stroke.**

At the end of the stroke, piston reaches the left end. Again during the next stroke, piston moves towards right, which creates partial vaccum in the left space of cylinder. Due to low pressure in the cylinder, delivery valve closes and suction valve opens and sucks liquid from sump for the next cycle.

Thus, suction and delivery strokes are alternately carried out and the liquid is pumped into the delivery tank during delivery stroke.